Derivation of NLL from a Bernoulli that uses a sigmoid logistic

$$\frac{\partial\, nll}{\partial(w^T x+b)} = \frac{\partial}{\partial a} nll = \frac{\partial}{\partial a}\left[y \log(p) + (1-y)\cdot\log(1-p)\right] = \frac{\partial}{\partial a} L$$

$$p \triangleq p(Y=y|X,\theta) = \sigma(w^T x + b) = \sigma(a) = \frac{1}{1+e^{-a}}$$

chain rule $\frac{dL}{da} = \frac{dL}{dp}\cdot\frac{dp}{da}$

- $\frac{dp}{da} = \frac{d}{da}\left(\frac{1}{1+e^{-a}}\right) = \frac{e^{-a}}{(1+e^{-a})^2} = \frac{e^{-a}+1-1}{(1+e^{-a})^2} = \frac{1+\cancel{e^{-a}}}{(1+e^{-a})^2} - \frac{1}{(1+e^{-a})^2}$ $\quad p \quad p$

$$= p(1-p)$$

- $\frac{dL}{dp} = \frac{\partial}{\partial p}\left(y\log(p) + (1-y)\log(1-p)\right) = \left(\frac{y}{p} - \frac{1-y}{1-p}\right)$

$$\frac{dL}{da} = \left(\frac{y}{p} - \frac{1-y}{1-p}\right)\cdot p(1-p) = p - y = \frac{\partial\, nll}{\partial(w^T x+b)}$$

To get $\frac{\partial}{\partial w} L$ and $\frac{\partial}{\partial b} L$,

chain rule $\frac{dL}{dp}\cdot\frac{\partial p}{\partial a}\cdot\frac{\partial a}{\partial w}$

- $\frac{da}{dw} = \frac{w^T x + b}{dw} = x$

$$\boxed{\frac{\partial L}{\partial w} = (p-y)x}$$

chain rule $\frac{dL}{dp}\cdot\frac{\partial p}{\partial a}\cdot\frac{\partial a}{\partial w}$

- $\frac{da}{dw} = \frac{w^T x + b}{dw} = 1$

$$\boxed{\frac{\partial L}{\partial b} = p(1-p)}$$